

Daniyar Omarov

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Employment

University of Alberta	Edmonton, AB, Canada
<i>Postdoctoral Fellow at the U. of Alberta with the Kantorovich Initiative (PIMS)</i>	<i>2024 – Present</i>

Education

Georgia Institute of Technology	Atlanta, GA, USA
<i>PhD in Mathematics, GPA: 3.94 out of 4.0</i>	<i>2018 – 2024</i>
Nazarbayev University	Astana, Kazakhstan
<i>Bachelor of Science in Mathematics, GPA: 3.9 out of 4.0</i>	<i>2014 – 2018</i>
University College London	Astana, Kazakhstan
<i>University Preparatory Certificate with Distinction</i>	<i>2013 – 2014</i>

Research Interests

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|-----------------------------|----------------------|
| ○ Optimal Transport Problem | ○ Numerical Analysis |
| ○ Dynamical Systems | ○ Optimization |

Publications and Preprints

- [4] Luca Nenna, Daniyar Omarov, and Brendan Pass. *Characterizing and computing solutions to regularized semi-discrete optimal transport via an ordinary differential equation*. 2025. arXiv: 2504.03030.
- [3] Luca Dieci and Daniyar Omarov. “Solving semi-discrete optimal transport problems: star shapedness and Newton’s method”. In: *Numerical Algorithms* (2024). doi: 10.1007/s11075-024-01903-y.
- [2] Luca Dieci and Daniyar Omarov. “Techniques for continuous optimal transport problem”. In: *Computers & Mathematics with Applications* 146 (2023), pp. 176–191.
- [1] Daniyar Omarov et al. “On the application of Sturm’s theorem to analysis of dynamic pull-in for a graphene-based MEMS model”. In: *Applied and Computational Mechanics* 12.1 (2018).

Poster Presentations and Talks

2024 Summer PIMS-CM2LA Kantorovich Initiative Retreat Meeting	August 2024
<i>University of British Columbia, Vancouver, BC, Canada</i>	
Gave a talk on “Solving Semi-Discrete OT: Star-shapedness and Newton’s method”	
Structural Dynamical Systems: Computational Aspects	June 2024
<i>Monopoli, Italy</i>	
With Prof. Luca Dieci, presented a minicourse on numerical methods in optimal transport	

SIAM Student Seminar <i>Georgia Institute of Technology, Atlanta, GA, USA</i> Gave a talk on "Solving Semi-Discrete Optimal Transport Problems"	April 2024
Florida-Georgia Applied Computational Math Student Workshop 2023 <i>Florida State University, Tallahassee, FL, USA</i> Gave a talk on "Techniques for continuous optimal transport problem"	April 2023
2023 Georgia Scientific Computing Symposium <i>Georgia State University, Atlanta, GA, USA</i> Presented a poster on "Techniques for continuous optimal transport problem"	February 2023
SIAM Gather Event for Applied Math Students <i>Georgia Institute of Technology, Atlanta, GA, USA</i> Gave a talk on "Numerical Algorithms for Optimal Transport Problem"	February 2021
Illinois Summer Research Symposium <i>University of Illinois Urbana-Champaign, Champaign, IL, USA</i> Presented a poster on "Zipf's Law: A Universal Law for Empirical Data from Word Frequencies to Olympic Records"	July 2016

Honors and Awards

Outstanding Teaching Assistant Award <i>School of Mathematics, Georgia Institute of Technology</i>	April 2023
Best Presentation Award <i>Florida-Georgia Applied Computational Math Student Workshop 2023</i>	April 2023
Certificate of Achievement for Outstanding Head TA <i>School of Mathematics, Georgia Institute of Technology</i>	Spring 2023
Certificate for Outstanding Efforts and Accomplishments <i>Society for Industrial and Applied Mathematics</i>	Spring 2022
Tech to Teaching Certificate in College Teaching <i>Center for Teaching and Learning, Georgia Institute of Technology</i>	Summer 2021
Certificate of Completion of REU Program <i>Department of Mathematics, University of Illinois Urbana-Champaign</i>	Summer 2016
Winner of "Taiburyl" Scholarship for Young Scientists and Researchers <i>Astana, Kazakhstan</i>	Fall 2015
Dean's List Awards for Excellent Academic Achievements <i>Department of Mathematics, Nazarbayev University</i>	2014 – 2018

Teaching Experience

University of Alberta	Edmonton, AB, Canada <i>Fall 2024</i>
○ Instructor: - <u>Fall 2024</u> - Math 201, <i>Differential Equations</i> - 123 students	

- **Instructor:** taught lectures and managed teaching assistants.
 - Fall 2022 - Math 1551, *Differential Calculus* - 101 students and 3 TA-s
 - Summer 2021 - Math 3406, *Second Course on Linear Algebra* - 24 students
- **Head Teaching Assistant:** taught recitations and managed teaching assistants.
 - Spring 2024 - Math 1554, *Linear Algebra* - 34 students and 13 TA-s
 - Fall 2023 - Math 1554, *Linear Algebra* - 36 students and 27 TA-s
 - Spring 2023 - Math 1554, *Linear Algebra* - 32 students and 8 TA-s
 - Fall 2020 - Math 2552, *Differential Equations* - 35 students and 12 TA-s
- **Lecture Assistant:** assisted the course instructor with lectures and held weekly office hours.
 - Spring 2021 - Math 3406, *Second Course on Linear Algebra* - 39 students
- **Teaching Assistant:** taught recitations and held weekly office hours.
 - Summer 2022 - Math 2552, *Differential Equations* - 33 students
 - Spring 2022 - Math 1554, *Linear Algebra* - 73 students
 - Fall 2021 - Math 1113, *Pre-Calculus* - 71 students
 - Spring 2021 - Math 2552, *Differential Equations* - 35 students
 - Summer 2020 - Math 1554, *Linear Algebra* - 38 students
 - Spring 2020 - Math 1554, *Linear Algebra* - 55 students
 - Fall 2019 - Math 2552, *Differential Equations* - 77 students
 - Summer 2019 - Math 4873/8873, *Computations in Dynamics* - 5 students
 - Spring 2019 - Math 1711, *Finite Mathematics* - 46 students
 - Fall 2018 - Math 2552, *Differential Equations* - 27 students

Extracurricular Activities

Georgia Tech SIAM Student Chapter

Atlanta, GA, USA

School of Mathematics, Georgia Institute of Technology

2019 – 2024

Treasurer (2020–2021): managed the funds of the chapter and organized social activities for students**President** (2021–2022): arranged student seminars and organized “Southeast SIAM Student Conference”

Georgia Tech High School Mathematics Competition

Atlanta, GA, USA

School of Mathematics, Georgia Institute of Technology

Spring 2019, Spring 2023

Helped to organize the competition and volunteered during the event

International Teaching Assistant (ITA) Liaison

Atlanta, GA, USA

Center for Teaching and Learning, Georgia Institute of Technology

Spring 2021

Designed learning materials for incoming ITA-s that address challenges of new experience

Nazarbayev University Mathematics Club

Astana, Kazakhstan

Department of Mathematics, Nazarbayev University

2016 – 2018

President (2016 - 2017): organized monthly seminars and conducted series of workshops in scientific computing platforms such as *Matlab* and *Mathematica*

Nazarbayev University Mentorship Program

School of Science and Technology, Nazarbayev University

Mentored two freshman students to help them with academic and social life

Astana, Kazakhstan

Spring 2016

Relevant Coursework

Georgia Institute of Technology

Atlanta, GA, USA

2018 – 2024

- Partial Differential Equations
- Advanced Linear Algebra
- Numerical Approximation Theory
- Multivariable Linear Systems and Control
- Numerical Methods for Dynamical Systems
- Optimal Transport Theory and Applications
- Introduction to Numerical Methods for PDE
- Iterative Methods for Systems of Equations

Nazarbayev University

Astana, Kazakhstan

2014 – 2018

- Introduction to Numerical Methods
- Mathematical Statistics
- Linear Programming
- Fourier Analysis

University of Illinois at Urbana-Champaign

Champaign, Illinois

Summer 2016

- Summer Workshop in Linear Algebra: Rational Canonical Form, Jordan Canonical Form, Orthogonal and Unitary Operators, and Spectral Theorem for Normal Operators
- Topics in Applied Probability: Random Walks, Ballot Theorem, Arcsine Law, and Betting Systems

Computer Skills

- Matlab - Advanced level
- Latex/Beamer - Advanced level
- Mathematica - Advanced level
- C++ - Intermediate level